Software Requirements Specification

SkyFly Airlines Reservation System

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1. Objective and Scope

The **Objective** of the Airlines Reservation System, "SkyFly Airlines Reservation System," is to provide a comprehensive and user-friendly platform for customers to book flight tickets and manage their reservations efficiently. The system aims to streamline the airline reservation process and enhance the overall user experience.

The **Scope** of the **SkyFly Airlines Reservation System** includes the following key features:

1. **Flight Search and Booking:** The system will allow users to search for available flights based on criteria such as departure city, destination, date, and passenger count. Users can then proceed to book their preferred flights.

2. **Seat Selection:** During the booking process, customers will have the option to select seats from an interactive seat map, subject to availability.

3. **Reservation Management:** Users will be able to view their existing flight reservations, make modifications to bookings (if allowed by the airline's policy), and cancel reservations if necessary.

4. **User Authentication and Profiles:** To access the booking functionalities, users will be required to create accounts and log in. The system will store user profiles containing essential information for future bookings.

5. **Payment Processing:** The Airlines Reservation System will integrate a secure payment gateway to facilitate online payments for flight bookings.

6. **Flight Availability and Schedule:** The system will provide real-time information on flight availability, schedules, and fares based on the selected travel criteria.

7. **Ticket Confirmation:** Upon successful booking, users will receive a booking confirmation containing essential details such as flight itinerary, ticket number, and payment receipt.

2. Project End Users

SkyFly Airlines Reservation System is designed to cater primarily to the following users:

1. **Customers**: Customers seeking to book flight tickets for their travel needs are the primarily end users of the system. Customers can use the application to search for available flights, select their preferred travel options, make reservations, and complete the booking process.

3. Features

3.1 User Authentication and Registration

Airlines Reservation System provides a user-friendly login interface for registered users to access their accounts. New users can signup to create a new account within the system. The Login and registration process involves the following:

Login: Each and every user should be authenticated with their credentials - User Name and Password to login into the system.

Signup: New Users can signup by providing essential details such as Name, Email Address, Contact Number, and creating Password.

3.2 Flights Menu

3.2.1 Searching for Flights:

Users can search for available flights based on their preferred departure and destination locations, along with the desired travel dates. The System will display a list of flights matching the search criteria.

3.2.2 Booking Flight:

Upon finding a suitable flight, users have the option to book the flight. The System will prompt the user to confirm their booking (Y or N). if confirmed it will be redirected to the booking page.

During the flight booking process, user will need to provide the following details:

i. **Seats count:** User should first start the booking process with the number of seats required.
ii. **Flight Id:** User should enter the desired flight id to proceed the booking.
iii. **Passenger Details:** Passenger should enter the basic details including Name, Age, Gender, Email, Address
iv. **Seats Selection:** User can select any class and any seat type and the ticket price is calculated accordingly based on the seats selection and class type.

3.2.3 Ticket Summary:

After successful flight booking, the system will generate the ticket summary that include all the booking details, such as flight information, passenger details, total ticket price, seat Allocated,

Upon finding a suitable flight, users have the option to book the flight. The System will prompt the user to confirm their booking (Y or N). if confirmed it will be redirected to the booking page.

3.2.4 View Ticket Itineraries:

Users with the booking id can view their booked ticket itineraries which contain comprehensive details about passenger name, flight number, airlines, booking id, travel details, seats booked and total price.

3.2.5 Explore Travel Options:

Users can explore travel options by information like departure city, destination city and airlines.

3.3 Special Services

Airlines Reservation System offers a range of special services to cater to passengers' specific needs and requirements. These services include:

a. Special Baggage

Users can request special baggage services to handle delicate or oversized baggage during their travel.

b. Special Meals On Board

Passengers with specific dietary requirements can request special meals to be served onboard during their flight.

c. Special Assistance at the Airport

The system provides options for passengers who require special assistance at the airport, such as wheelchair assistance or help for passengers with reduced mobility.

d. Travelers with Reduced Mobility or Needing a Wheelchair

Passengers who have reduced mobility or require a wheelchair during their journey can make the necessary arrangements through the system.

e. Travelers Traveling with Animals

Users can specify if they are traveling with animals or pets and request any additional assistance or facilities required for their pets during the flight.

3.4 Ticket Cancellation

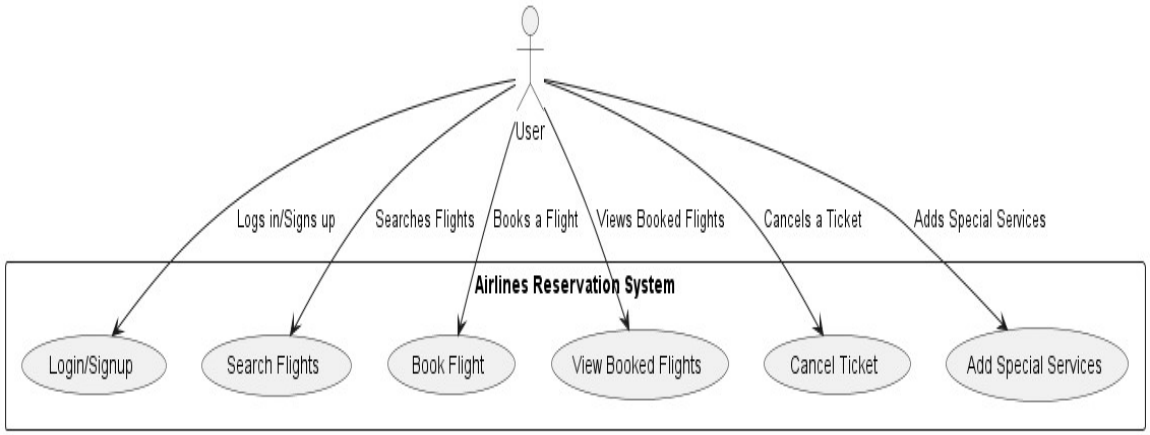
The System shall prompt users to enter the booking id for cancellation. The system will be verifying the booking id and display ticket details for verification. The system shall allow users to provide an optional cancellation reason, and then it will calculate the refund amount based on the total ticket price, deducting 20% as per privacy policies. The system shall update the booking status to "cancelled" and store the cancellation reason, if provided.

3.5 Exit

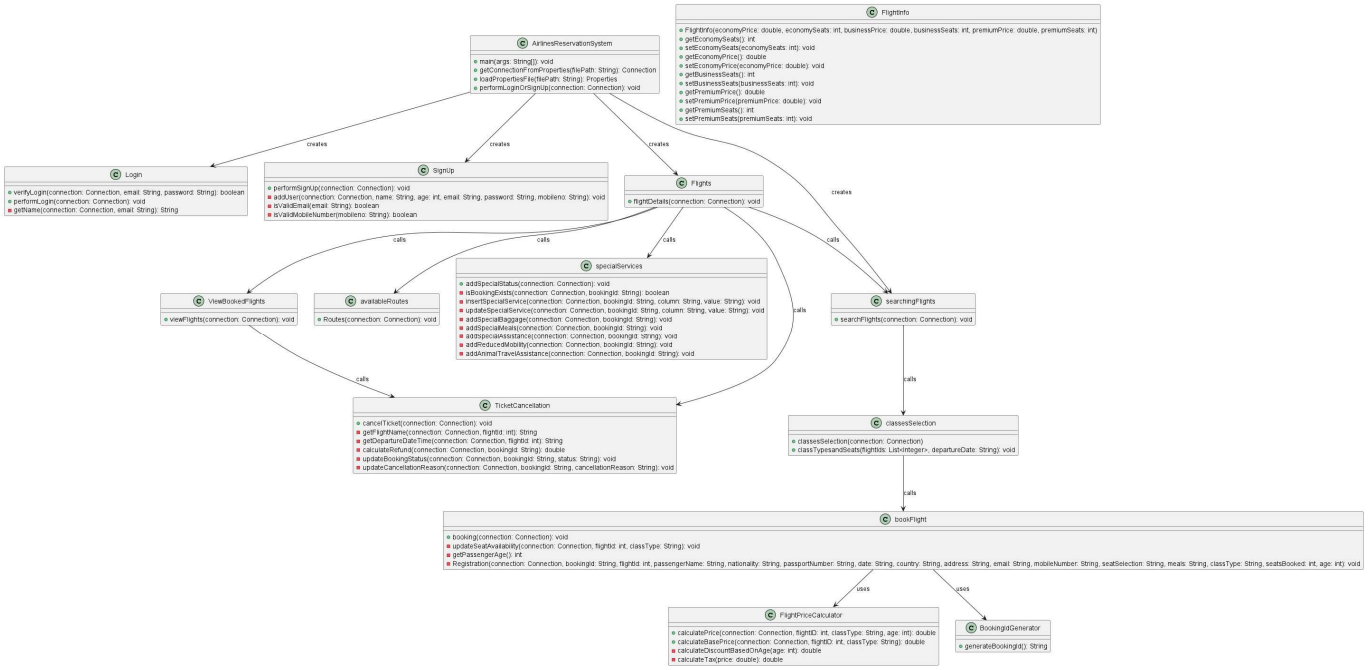
Once the user is done with the process, he can exit the application.

4. Diagrams

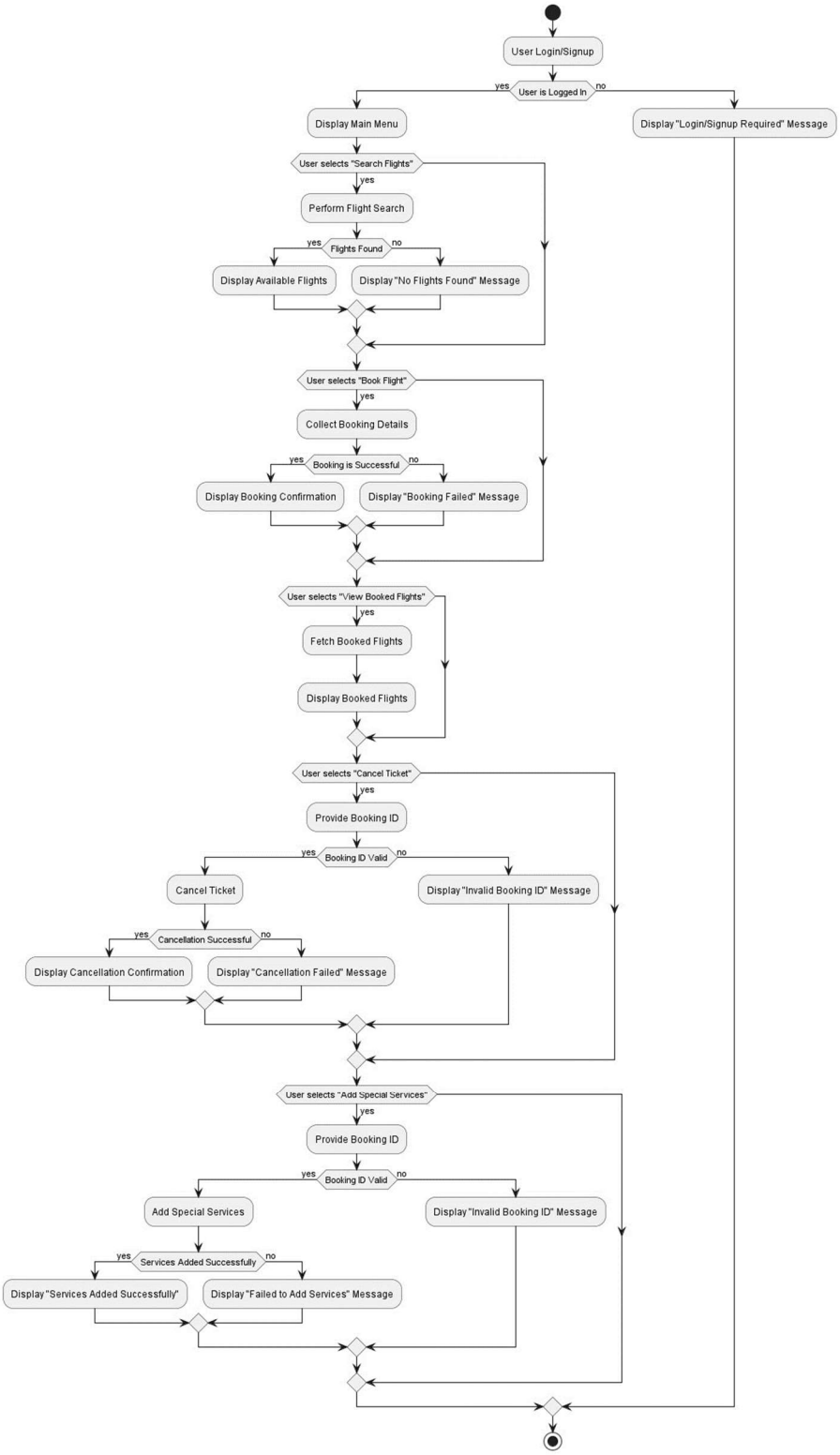
4.1Use Case Diagram



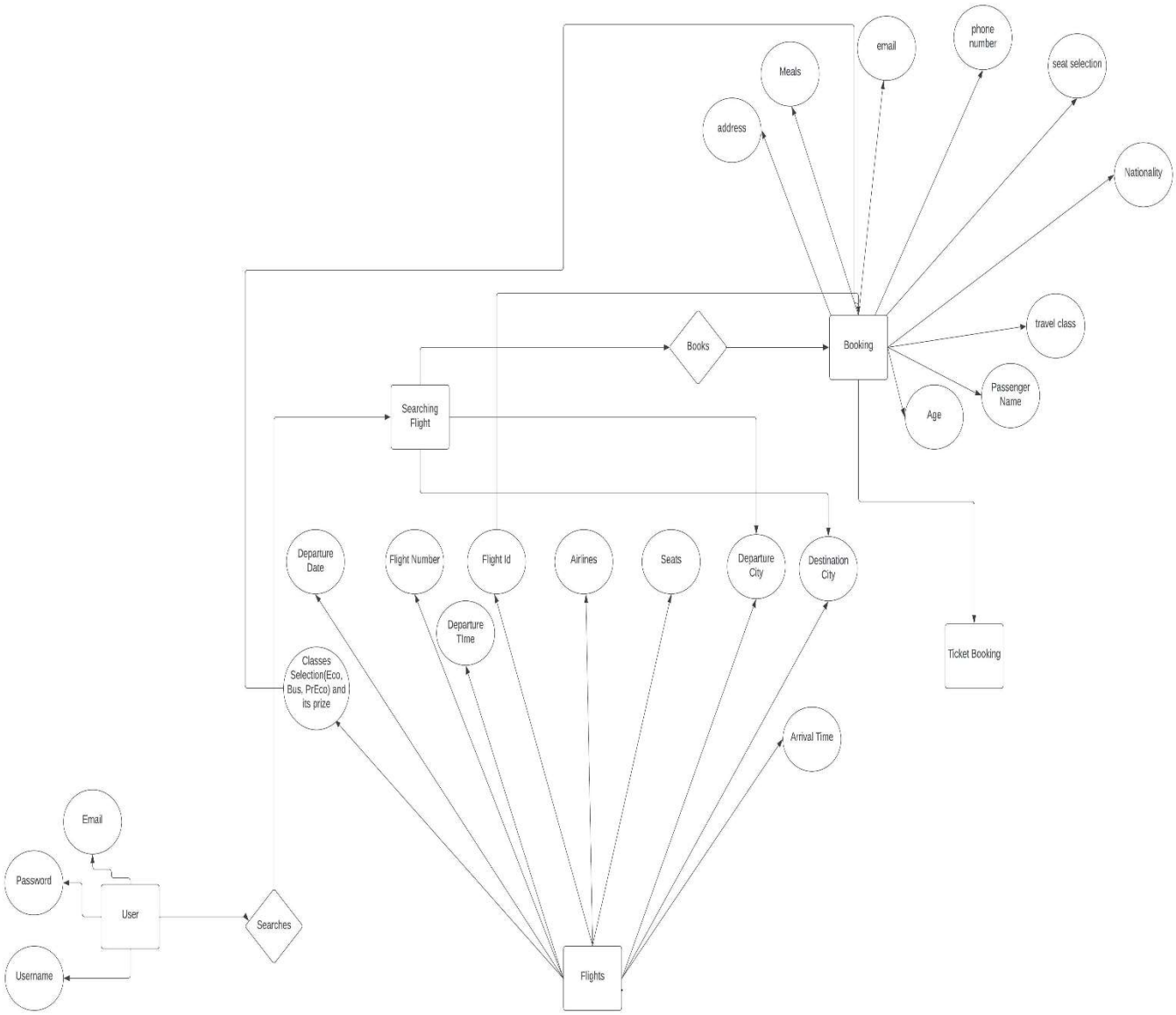
4.2 Class Diagram



4.3 Flow Chart Diagram



4.4. ER Diagram



5. Functional Requirements

1. The user must be able to view available flights and their details.

2. The user should be able to search for flights based on departure and destination cities.

3. The user should be able to select a flight and book tickets for a specific class type (Economy, Business, Premium).

4. The system should calculate the total price of the tickets based on the class type and other factors like age, meal preferences, etc.

5. The user should be able to provide passenger details (name, nationality, passport number, etc.) during the booking process.

6. The system should generate a unique booking ID for each successful booking.

7. The user should be able to view their booked tickets and their details using the booking ID.

8. The user should have the option to cancel their booked tickets and receive a refund based on the cancellation policy.

9. The system should send a confirmation email to the user after successful booking.

10. The system should provide a login/signup feature to manage user accounts and bookings.

6. Non-Functional Requirements

1. The console application should have a user-friendly interface for easy navigation.

2. The application should handle errors gracefully and provide informative error messages.

3. The system should be able to handle multiple concurrent user requests without performance degradation.

4. The data stored in the database should be secure and protected from unauthorized access.

5. The application should be well-tested to ensure it is bug-free and reliable.

6. The system should be scalable to accommodate future updates and enhancements.

7. The console application should have proper documentation for ease of understanding and maintenance.

8. The system should maintain a good level of code quality and follow coding standards.

9. The application should be regularly maintained to ensure it is up-to-date and functioning properly.

10. The console application should be platform-independent and compatible with different operating systems.

7. Reports

This system has reporting capabilities. It shows the report of total delivered, total bounced mails, total deferred and total rejected. It also displays each mails and its status separately and also stored in the database. So, it is useful to know each mail status.